

Research Publications

Institute of Computer Science

June 14, 2026

- [1] A. B. Doria and V. R. M. Madrid. Developing a high-throughput phenotyping software for eggplant through trichome analysis using deep learning. In *Proceedings of the 3rd International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2026)*, Boracay, Philippines, February 2026. Conference Paper, Scopus-Indexed.
- [2] Q. Z. L. Ramos and A. B. Doria. Vanishingpoint: A virtual gallery web application for interactive portfolio creation. In *Proceedings of the 3rd International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2026)*, Boracay, Philippines, February 2026. Conference Paper, Scopus-Indexed.
- [3] Marveen Bernabe and Val Randolph M. Madrid. Mambo (multi ai model bench operations): A framework for dynamic model switching and benchmarking on-device ai. In *Proceedings of the 26th Philippine Computing Science Congress (PCSC 2026)*, University of the Immaculate Conception, Davao City, Philippines, April 2026. Conference Paper.
- [4] Angelo Arenas and Val Randolph M. Madrid. Judo throw recognition using computer vision techniques. In *Proceedings of the 26th Philippine Computing Science Congress (PCSC 2026)*, University of the Immaculate Conception, Davao City, Philippines, April 2026. Conference Paper.
- [5] Trisha Jean Ramos and Rachel Editra O. Roxas. Tudlo: Parameter-efficient fine-tuning for tagalog and cebuano early-grade mathematics explanation generation. In *Proceedings of the 26th Philippine Computing Science Congress (PCSC 2026)*, University of the Immaculate Conception, Davao City, Philippines, April 2026. Conference Paper.
- [6] S. A. S. Dela Cruz and M. A. J. B. Regis. Birdview: Human crowd density visualization tool from video feed to floor plan using deep learning. In *Proceedings of the 26th Philippine Computing Science Congress (PCSC 2026)*, University of the Immaculate Conception, Davao City, Philippines, April 2026. Conference Paper.
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 - [10] A. M. A. Velasquez and R. N. C. Recario. Comparative machine learning and explainable ai for predicting coronary artery disease. In *2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, pages 1–6, Boracay Island, Philippines, 2026. Conference Paper, Scopus-Indexed.
 - [11] A. R. B. Ramillano and R. N. C. Recario. Learnit: Ai-powered learning companion in education setting using text segmentation and contextual condensing (tscc). In *2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, pages 1–6, Boracay Island, Philippines, 2026. Conference Paper, Scopus-Indexed.
 - [12] A. C. C. Maaño, D. J. Mercado, and R. N. C. Recario. Weighted filtering approaches for a filipino movie recommendation system. In *2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, pages 1–6, Boracay Island, Philippines, 2026. Conference Paper, Scopus-Indexed.
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 - [14] R. T. Santos, A. J. Jacildo, A. D. Montecillo, R. J. D. Flores, and M. P. A. De Leon. Cross-sectional analysis of bacterial co-occurrence networks of bat gut-associated microbiomes sampled across various cave sites in calabarzon, philippines. Poster paper presented at IWOMB 2025 - International Workshop on Mathematical Biology, January 2025. Poster Paper.
 - [15] Zenith Arnejo, Benoit Gaudou, Mehdi Saqalli, and Nathaniel Bantayan. Luntian: An agent-based model of an industrial tree plantation for promoting sustainable harvesting in the philippines. *Forests*, 16(8):1293, 2025. Journal Paper.

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- [24] J. P. Pabico. Regulating ai in education and empowering educators: Policy, governance, and ethical considerations (invited plenary paper). In *1st Research Conference on Mission-Driven Education (RCMDE 2024)*, Universidad de Sta. Isabel de Naga, Naga City, July 2024. Conference Paper.
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